between the instrument cluster and the transfer case switch as required.

INDICATOR STAYS ILLUMINATED WITH 4WD MODE NOT SELECTED

(1) Disconnect and isolate the battery negative cable. Disconnect the engine wire harness connector for the transfer case switch from the transfer case switch connector receptacle. Check for continuity between the ground circuit and the 4WD switch sense circuit terminals in the transfer case switch connector receptacle. There should be no continuity. If OK, go to Step 2. If not OK, replace the faulty transfer case switch.

(2) Remove the instrument cluster from the instrument panel. Disconnect the instrument panel wire harness connector (Connector C2) for the instrument cluster from the connector receptacle on the back of the cluster housing. Check for continuity between the 4WD switch sense circuit cavity of the instrument panel wire harness connector (Connector C2) for the instrument cluster and a good ground. There should be no continuity. If OK, replace the faulty instrument cluster. If not OK, repair the shorted 4WD switch sense circuit between the transfer case switch and the instrument cluster as required.

SKIS INDICATOR

DESCRIPTION

Fig. 24 SKIS Indicator

A Sentry Key Immobilizer System (SKIS) indicator is standard equipment on all instrument clusters, but is only functional on vehicles equipped with the optional SKIS (Fig. 24). The SKIS indicator is located near the upper edge of the instrument cluster overlay, to the left of center. The SKIS indicator consists of a stencil-like cutout of a graphical representation or icon of a key that is circled and crossed-out in the opaque layer of the instrument cluster overlay. The dark outer layer of the overlay prevents the indicator from being clearly visible when it is not illuminated. A amber Light Emitting Diode (LED) behind the cutout in the opaque layer of the overlay causes the indicator to appear in amber through the translucent outer layer of the overlay when the indicator is illuminated from behind by the LED, which is soldered onto the instrument cluster electronic circuit board. The SKIS indicator is serviced as a unit with the instrument cluster.

OPERATION

The Sentry Key Immobilizer System (SKIS) indicator gives an indication to the vehicle operator of the status of the SKIS. This indicator is controlled by the instrument cluster circuit board based upon electronic messages received by the cluster from the Sentry Key Immobilizer Module (SKIM) over the Programmable Communications Interface (PCI) data bus. The SKIS indicator Light Emitting Diode (LED) is completely controlled by the instrument cluster logic circuit, and that logic will only allow this indicator to operate when the instrument cluster receives a battery current input on the fused ignition switch output (run-start) circuit. Therefore, the LED will always be off when the ignition switch is in any position except On or Start. The LED only illuminates when it is provided a path to ground by the instrument cluster transistor. The instrument cluster will turn on the SKIS indicator for the following reasons:

- **Bulb Test** - Each time the ignition switch is turned to the On position, the SKIM sends a message to the cluster to illuminate the SKIS indicator for about three seconds as a bulb test.
- **SKIM Lamp-On Message** - Each time the cluster receives a lamp-on message from the SKIM, the SKIS indicator will be illuminated. The indicator can be flashed on and off, or illuminated solid, as dictated by the SKIM message. For more information on the SKIS and the SKIS indicator control parameters, (Refer to 8 - ELECTRICAL/VEHICLE THEFT SECURITY - OPERATION). The indicator remains illuminated until the cluster receives a lamp-off message from the SKIM or until the ignition switch is turned to the Off position, whichever occurs first.
- **Actuator Test** - Each time the cluster is put through the actuator test, the SKIS indicator will be turned on, then off again during the bulb check portion of the test to confirm the functionality of the LED and the cluster control circuitry.

The SKIM performs a self-test each time the ignition switch is turned to the On position to decide whether the system is in good operating condition. The SKIM then sends the proper SKIS lamp-on or lamp-off messages to the instrument cluster. For further diagnosis of the SKIS indicator or the instrument cluster circuitry that controls the indicator, (Refer to 8 - ELECTRICAL/INSTRUMENT CLUSTER - DIAGNOSIS AND TESTING). If the instrument cluster turns on the SKIS indicator after the bulb test, either solid or flashing, it indicates that a SKIS malfunction has occurred or that the SKIS is inoperative. For proper diagnosis of the SKIS, the PCI data bus, or the electronic message inputs to the instrument cluster that control the SKIS indicator, a DRBIII® scan tool is required. Refer to the appropriate diagnostic information.